

Qatar-1b

R-0635

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230625 · created 2026-07-25T17:31:11+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460120.7057 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.013
Transit depth (Rp/Rs)^2	3.24 +/- 0.48 [%]
Orbital Inclination (inc)	83.85 +/- 0.89
Airmass coefficient 1 (a1)	0.922 +/- 0.012
Airmass coefficient 2 (a2)	0.055 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [217, 266]
Optimal Aperture	3.31
Optimal Annulus	22.67
Transit Duration (day)	0.0705 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.18 $\pm$ 0.013 vs 0.1463 (deviation 2.59 σ)
Duration fit vs published	0.0705 d vs 0.0692 d (deviation 0.25 σ)
Mid-time O-C	-6.2 min
Depth SNR	13.8
Residual scatter	1.33 %

Light curve

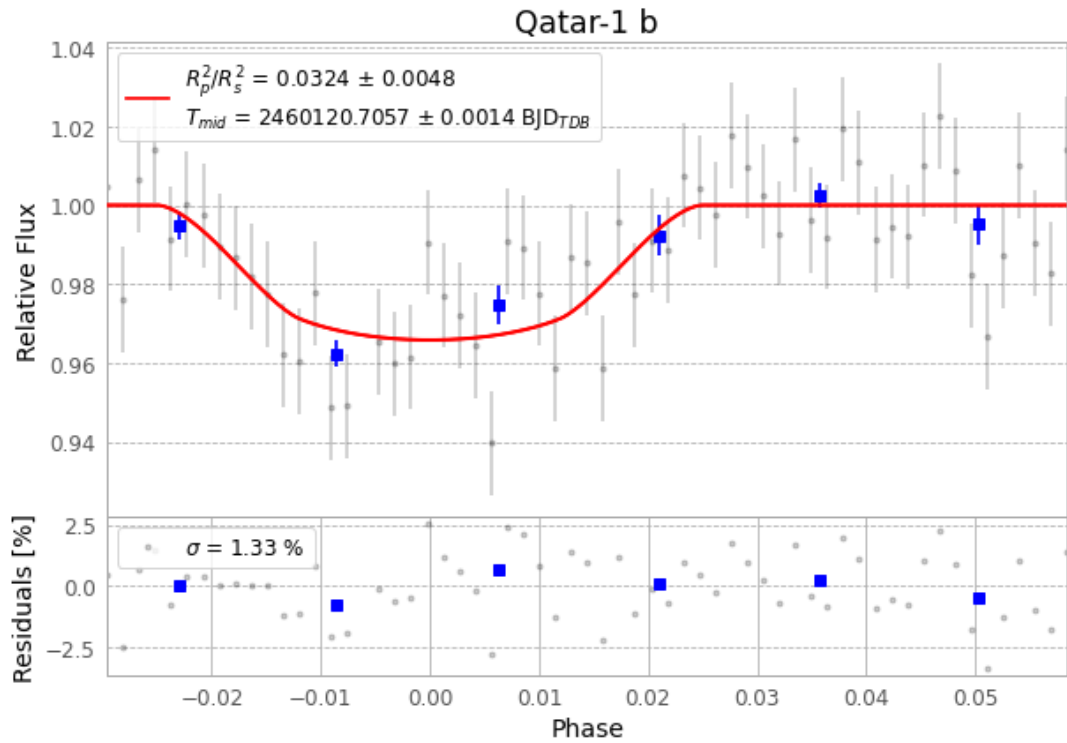


Figure 1 — FinalLightCurve_Qatar-1 b_2023-06-24.png (SHA-256 640f947b1f04b3c4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 244], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b500ed6a350a8466...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

61 FITS frames (40 MB), Qatar-1230625035415.FITS → Qatar-1230625065415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230625.log (45bb43a45319...), FinalLightCurve_Qatar-1 b_2023-06-24.png (640f947b1f04...), FinalLightCurve_Qatar-1 b_2023-06-24.csv (bfb8e68f2b0d...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 17:31Z.